

Siddharth S. Sane

+1 (672)-667-6841 | siddharthsane2@gmail.com | Element200.github.io

EDUCATION

Yale University

PhD Physics. CGPA: ???

- Thesis title: ???
- Supervisor: ???

New Haven, CT (USA)

2026 – Present

Simon Fraser University

M.Sc. Physics. CGPA: 4.29/4.33

- Thesis title: *Anomalous cooling in systems quenched at finite rates*
- Supervisor: Dr. John Bechhoefer

Burnaby, BC (Canada)

2023 – Present

Azim Premji University

B.Sc. Physics (with honours). CGPA: 9.20/10

- Thesis title: *Deformable liquid crystal droplets in surfactant media*
- Supervisor: Dr. Rema Krishnaswamy

Bengaluru, KA (India)

2020 – 2023

Delhi Public School, Bangalore North

Grade XII (Central Board of Secondary Education). Final average grade: 95.4%

Bengaluru, KA (India)

2019 – 2020

Delhi Public School, Bangalore North

Grade X (Central Board of Secondary Education). Final average grade: 90.8%

Bengaluru, KA (India)

2017 – 2018

PUBLICATIONS

¹M. Kumar, S. **Sane**, A. Murali, and S. Thutupalli, “Temperature switchable self-propulsion activity of liquid crystalline microdroplets”, *Soft Matter* **21**, 3782–3788 (2025).

²T. Deora, S. S. **Sane**, and S. P. Sane, “Wings and halteres act as coupled dual oscillators in flies”, *eLife* **10**, e53824 (2021).

EXPERIENCE

Research Assistant

Simon Fraser University

2023 – 2026

Burnaby, BC (Canada)

- Conducted research on the Mpemba effect under the guidance of Prof. John Bechhoefer.
- Experimental methods involve specialised optical trap that uses feedback to dynamically adjust position of trap based on measured position of Brownian particle, allowing for arbitrary time-dependent potentials on particles
- Developed mini-library to efficiently simulate and analyse the dynamics of Brownian particles in Python using stochastic integration of the Langevin equation

Teaching Assistant

Simon Fraser University

Spring 2025; Spring 2026

Burnaby, BC (Canada)

- Lab Assistant for PHYS234 (undergraduate physics laboratory) for Prof. John Bechhoefer.
- Assisted about 25 students with experiments, helped debug their code, tested experiments beforehand and assisted in updating lab protocols.
- Graded lab notebooks and exams, and provide feedback.
- Won a student-nominated award for excellence in teaching in 2025.

Teaching Assistant

Simon Fraser University

Spring 2024

Burnaby, BC (Canada)

- Teaching Assistant for PHYS100 (pre-calculus physics) for Prof. Erol Girt and Dr. Simin Bagheri-Najmi.
- Conducted weekly tutorials for five classes of 20 students each.
- Advised students on problem solving techniques.
- Graded student homework and exams.

Intern (Research Education Advancement Programme)

National Centre for Biological Sciences

2022 – 2023

Bengaluru, KA (India)

- Third year of REAP involved a year-long part-time internship in a research laboratory.
- Conducted experiments with Dr. Manoj Kumar, a postdoctoral fellow in the lab of Prof. Shashi Thutupalli.
- Conducted experiments on self-propelling liquid crystal (8CB) droplets under various temperatures and surfactant concentrations, particularly studying the onset of self-propulsion under smectic-nematic phase transitions.
- Experiments involved techniques including microfluidics (for production of fixed-radius droplets), inverted polarisation and bright-field microscopy, and particle image velocimetry.
- Research contributed to a published article in *Soft Matter* [1].

Undergraduate researcher

2022 – 2023

Azim Premji University

Bengaluru, KA (India)

- Honours degree involved research contributing to a thesis under the guidance of Prof. Rema Krishnaswamy.
- Conducted experiments on self propelling, deforming large liquid crystal (5CB) droplets in surfactant media.
- Built heating stage for inverted microscope.
- Conducted deformation analysis on large droplets using Python, including some AI segmentation tools.
- Also learnt about low-Reynolds number hydrodynamics and nematohydrodynamics from de Gennes and Prost's textbook as a guided reading project under Prof. Krishnaswamy.

Student

2020 – 2022

Jawaharlal Nehru Planetarium (remote)

Bengaluru, KA (India)

- REAP is a three-year programme for students of physics interested in research, with two years of coursework and one year of internship.
- Attended and completed assignments and exams for uncredited courses on advanced mechanics, mathematical methods in physics, variational calculus, electrodynamics, quantum mechanics, astronomy, and thermodynamics, conducted by researchers from the Indian Institute of Science, International Centre for Theoretical Sciences, Raman Research Institute, Jawaharlal Nehru Planetarium, Azim Premji University.
- Won scholarships in all three years of REAP.
- Won an award for excellence in the programme.

Guided reading project

2021

International Centre for Theoretical Sciences (remote)

Bengaluru, KA (India)

- Learned about soft matter physics from Selinger's textbook under the guidance of Prof. Vijay Krishnamurthy.

Intern

2020

National Centre for Biological Sciences (remote)

Bengaluru, KA (India)

- Digitised Dr. Tanvi Deora's high speed videos of tethered flies flapping.
- Wrote code to validate data, perform error analysis, etc.
- Trained DLTdv8, an open source neural network for MATLAB, to identify parts of fly wings.
- Work contributed to a published paper with Dr. Deora and Prof. Sanjay Sane in *eLife* [2].

SCHOLARSHIPS

Graduate fellowship award

2025

Department of Physics, Simon Fraser University

Burnaby, BC (Canada)

Special graduate entrance scholarship

2023

Department of Physics, Simon Fraser University

Burnaby, BC (Canada)

REAP scholarship

2021, 2022, 2023

Jawaharlal Nehru Planetarium

Bengaluru, KA (India)

AWARDS AND ACHIEVEMENTS

Physics TA teaching award

2025

Department of Physics, Simon Fraser University

Burnaby, BC (Canada)

- Awarded after student nomination for excellence in teaching and lab assistantship.

Travel and research award

2024

Department of Physics, Simon Fraser University

Burnaby, BC (Canada)

C. V. Visveswara award

2023

Jawaharlal Nehru Planetarium

Bengaluru, KA (India)

- Awarded for excellence in the Research Education Advancement Programme.

Subject topper certificate, mathematics

2020

Delhi Public School, Bangalore North

Bengaluru, KA (India)

- Awarded for receiving the highest marks in mathematics in my class of about 250 students.

Certificate of merit

2020

Delhi Public School, Bangalore North

Bengaluru, KA (India)

- Awarded for receiving over 95% in four of the five science stream subjects.

Anomalous cooling in systems quenched at finite rates

SAFETY TRAINING AND CERTIFICATIONS

Radiation safety training (sealed sources and portable gauges)

Dec 2024

Simon Fraser University (online)

Burnaby, BC (Canada)

Student machine shop course

Jun 2024

Simon Fraser University

Burnaby, BC (Canada)

Laboratory safety training (including WHMIS 2015)

Feb 2024

Simon Fraser University (hybrid)

Burnaby, BC (Canada)

EHS laser safety training

Dec 2023

Simon Fraser University (online)

Burnaby, BC (Canada)

SFU safety essentials (including SFU safety orientation)

Dec 2023

Simon Fraser University (online)

Burnaby, BC (Canada)

EXTRACURRICULAR ACTIVITIES

Union steward, department of physics

2025–2026

Teaching and Support Staff Union

Burnaby, BC (Canada)

- Acted as the liaison between students from the department and the TA and RA union.

Graduate Student Society student representative, dept. of physics

2024–2025

Simon Fraser University

Burnaby, BC (Canada)

- Represented the department of physics in the GSS caucus, voting on various decisions related to healthcare, etc.

Organiser, Frontiers in Biophysics

2024

Simon Fraser University

Burnaby, BC (Canada)

- Helped organise a student-run conference for biophysicists from across the pacific northwest.

Organiser, L^AT_EX workshop, physics club

2023

Azim Premji University

Bengaluru, KA (India)

- Organised a beginner's workshop on behalf of the physics club on the use of the typesetting language L^AT_EX for beginners. This workshop was the most attended event the physics club had ever conducted thus far.

Contestant, CBSE national science competition

2017

O.P. Jindal Global School

New Delhi, Delhi (India)

- Represented the state of KA at a science competition run by the Central Board of Secondary Education (CBSE) for high schoolers, after clearing the state level competition in Bengaluru.

Contestant, TechFest Robowars

2014

Indian Institute of Technology, Bombay

Mumbai, MH (India)

- I jointly built a combat robot with a friend to fight in a contest. We were the youngest contestants at the event.

TECHNICAL SKILLS

Experimental skills: Optical trapping techniques, electronics, microfluidics, brightfield, fluorescence, and polarisation microscopy on an inverted microscope. Certified for use of heavy machinery including lathes, mills, etc., high-intensity lasers, and work with hazardous chemicals.

Software: proficient in Python, L^AT_EX, C++ based languages like Arduino (for microcontrollers) and OpenSCAD (for 3D printing); can use Julia, Mathematica, C, Igor Pro, LabVIEW, etc. Know enough about programming languages to quickly pick up new ones. Familiar with standard Microsoft/ Google/ OpenOffice word processors, spreadsheets, etc. Familiar with (but not reliant on!) the use of large language models and various AI tools.

Libraries: Standard data analysis, image analysis, and optimisation libraries for Python and Julia: Pandas, NumPy, Matplotlib, OpenCV, Xarray, Numba, etc. Familiar with LabVIEW's FPGA programming libraries.